

Enhancing Productivity Applications for People who are Blind using AI Assistants

Minoli Perera
Monash University
Melbourne, Victoria, Australia
minoli.perera@monash.edu

ABSTRACT

Productivity applications, such as word processors, spreadsheets and presentations, have become indispensable tools in the workplace, higher education, and personal settings. These applications are primarily accessed by blind users through the use of screen readers, which face numerous accessibility and usability challenges that hinder productivity and independence. My research aims to understand the severity of these challenges, explore the design space for potential AI-based Assistants, and propose design guidelines for more accessible and usable applications. The research employs a Design Thinking and Co-design approach, with the active involvement of blind users throughout the project. I have conducted a survey and a user study to gain insights into blind users' experiences with productivity applications. The next phase of the project will delve into AI assistant capabilities and interaction techniques aimed at enhancing blind users' productivity and independence.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in accessibility**; *Accessibility technologies*.

KEYWORDS

blind, accessibility, productivity applications, assistive technology, screen readers, generative AI, AI assistants, virtual assistants, voice assistants

ACM Reference Format:

Minoli Perera. 2024. Enhancing Productivity Applications for People who are Blind using AI Assistants. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '24)*, May 11–16, 2024, Honolulu, HI, USA. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3613905.3638180>

1 RESEARCH STATUS

I am a second-year PhD student within the Faculty of Information Technology at Monash University, Australia. I have completed my PhD confirmation milestone and am currently at the halfway point of my expected 42-month candidature. Typically, universities in Australia have a 3.5 year PhD process for Computer Science. I have

conducted two studies to understand the problem space and have planned for two more to explore the design space.

2 BACKGROUND AND MOTIVATION

Productivity applications, such as word processors, spreadsheets, and presentations, play an important part in workplace and higher education settings. These tools are included in productivity suites such as Microsoft 365 [33], Google Workspace [20] and Apple iWork [4], and they are widely used around the world, having become a necessity for performing work-related as well as personal tasks [29]. The user interfaces of the majority of these applications are primarily created for sighted user interactions and heavily depend on visual cues [28].

Productivity applications and assistive technologies such as screen readers must collaborate to enable access for blind users to applications. Screen readers render content on a screen in synthetic speech and Braille output [50]. Due to their non-visual interaction and reliance on auditory feedback, blind users encounter many accessibility and usability challenges when working with these applications. They face difficulties understanding visual information [22, 23, 25, 47], obtaining overview information [50, 51], finding the desired content [36], understanding and applying formatting in documents [16, 35], and working in collaborative environments [9, 13, 14]. Apart from learning the functionalities of the applications, blind users need to acquire skills for effectively working with them using assistive technologies.

Previous research has revealed that blind employees often need extra time and effort to accomplish tasks when compared to their sighted colleagues and often depend on sighted assistance [10, 49]. It has been reported that they need to “work twice as hard” to achieve the same output, job performance, recognition, and pay in contrast to their sighted colleagues [41]. Several studies have been undertaken to enhance the accessibility and usability of productivity applications through feature additions [27], interface customizations [36, 48] and multi-modal systems [53, 54]. However, despite the potential benefits of these solutions, the majority of them have not made their way into mainstream products. Our work, along with previous studies, has identified a pressing need to address the challenges and enhance the experience for blind users.

In recent years, virtual assistants such as Siri [5], Alexa [3] and Google Assistant [19] have become popular and are utilized by many blind and sighted people. These assistants are currently used for more lightweight and playful tasks such as playing music, determining weather [30, 31] and are limited to productivity related tasks such as accessing emails, scheduling meetings, and setting reminders [2]. Prior research [1, 2, 11, 43, 46] that focused on the usability of virtual assistants for blind users has revealed that, despite

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).
CHI EA '24, May 11–16, 2024, Honolulu, HI, USA
© 2024 Copyright held by the owner/author(s).
ACM ISBN 979-8-4007-0331-7/24/05
<https://doi.org/10.1145/3613905.3638180>

the challenges primarily encountered around speech recognition errors and an inability to follow up on tasks, the use of virtual assistants ultimately proved beneficial. They offer more efficient interactions compared to navigating desktop or mobile interfaces using screen readers and provide more affordable and natural interactions [2]. However research in this space concerning blind users has primarily focused on home-based tasks, gaining understanding about their surroundings [17, 38, 39] and improving web accessibility [7, 8, 24, 44]. Although blind users have expressed an interest in utilizing virtual assistants in work settings [2], there has been limited investigation into this technology in the domain of productivity-related tasks [18, 26, 52].

While more advanced large language model based chatbots, such as ChatGPT [40] and Bard [21], have been recently introduced, holding significant potential to address current issues and improve productivity applications for blind users, they have not yet been explored in detail. We believe that investigating the role of these generative conversational AI-based assistants in a productivity-oriented context is not only timely but also highly significant, as it has the substantial potential to increase productivity and independence for blind users. This is the area in which I position my research.

3 RESEARCH QUESTIONS

The aim of the research is to explore how recent advancements in generative conversational AI can be leveraged to enhance the experience of blind users with productivity applications. To achieve this, we pose three questions:

RQ1: What challenges do blind users currently face when using productivity applications, and what is the potential usefulness of AI assistants in such applications?

This question aims to explore the accessibility and usability issues faced by blind users when working with productivity applications, understand their implications, and prioritize areas and applications that require more attention. It also explores blind users' preferences regarding the potential usefulness of AI assistance in productivity applications.

RQ2: In which context can generative conversational AI-based technologies assist blind users when working with productivity applications?

The objective of this question is to understand the expectations and needs of blind users regarding assistant capabilities and the desired interaction techniques for efficient task performance. While a recent study [42] has explored audio-based conversational assistant behaviors for screen reader users, primarily in the context of the web, this question delves into the use of different input and output modalities, such as text or voice commands, which would be more suitable within a productivity-oriented setting. Productivity applications involve more complex interactions such as authoring, editing, formatting and collaborating compared to the web, which is primarily focused on searching, browsing, and form filling. Co-design will provide a more nuanced understanding of how blind users naturally prefer to use these AI assistants in work related tasks.

RQ3: How effective are AI assistants in enhancing the experience of blind users with productivity applications?

Building on the insights gained from RQ2, this question will investigate and evaluate the user interactions and develop design guidelines for efficient and effective AI-assisted interactions with productivity applications. The guidelines will outline the optimal ways to utilize AI assistance to enhance the accessibility and usability of these applications for blind users.

4 RESEARCH APPROACH

Studies with the blind users are essential to get more insights on the technology based challenges and to understand their specific needs when working with productivity applications. Thus, a more user-centred approach, engaging users from the start to end of the project is essential for its success. Furthermore, people who are blind or have low vision (BLV) believe that greater involvement in research design and practice, along with providing them with more control over research activities, would yield greater contributions to the accessibility field [15]. Therefore, we have chosen a combination of **Design Thinking** [37] and **Co-design** [45] approaches, which will enable us to keep the users in the loop throughout the project. This will enable us to address users' needs and preferences in an iterative, non-linear manner throughout the process. This approach is gaining attention in accessibility research related to solving problems for BLV individuals [32]. Figure 1 illustrates the methodological approach of the project aligned with the research questions.

5 CURRENT PROGRESS

This section outlines the work conducted to address RQ1. Based on the Design Thinking approach (Figure 1), the first stage is to empathise with the user to know their actual challenges and needs. As the first step of the project, I learned how to use screen readers and access productivity applications non-visually, aiming to gain a better understanding of the blind user's experience. Next, we conducted the following two studies to explore the problem space.

5.1 Study One: Survey with blind users to understand their experience with productivity applications

An online survey was conducted to understand the severity of challenges in productivity applications and their resultant impact on productivity and independence. Participants were surveyed about application use, accessibility and usability challenges, as well as suggestions and preferences for more usable applications. The survey also explored the perceived usefulness of an AI-based assistant in productivity applications. The survey was screen reader accessible and consisted of 22 questions, including multiple-choice, Likert-scale, and open-ended questions.

In the first half of the survey, we collected users' demographic information, assessed their comfort level with technology, explored their use of productivity applications, and inquired about their learning methods. The applications were derived from an examination of existing productivity suites [4, 20, 33]. Then, we posed open-ended questions to elicit information about the worst accessibility issues, the applications in which these issues occur, and suggestions to address these issues. We also inquired about the impact of these issues

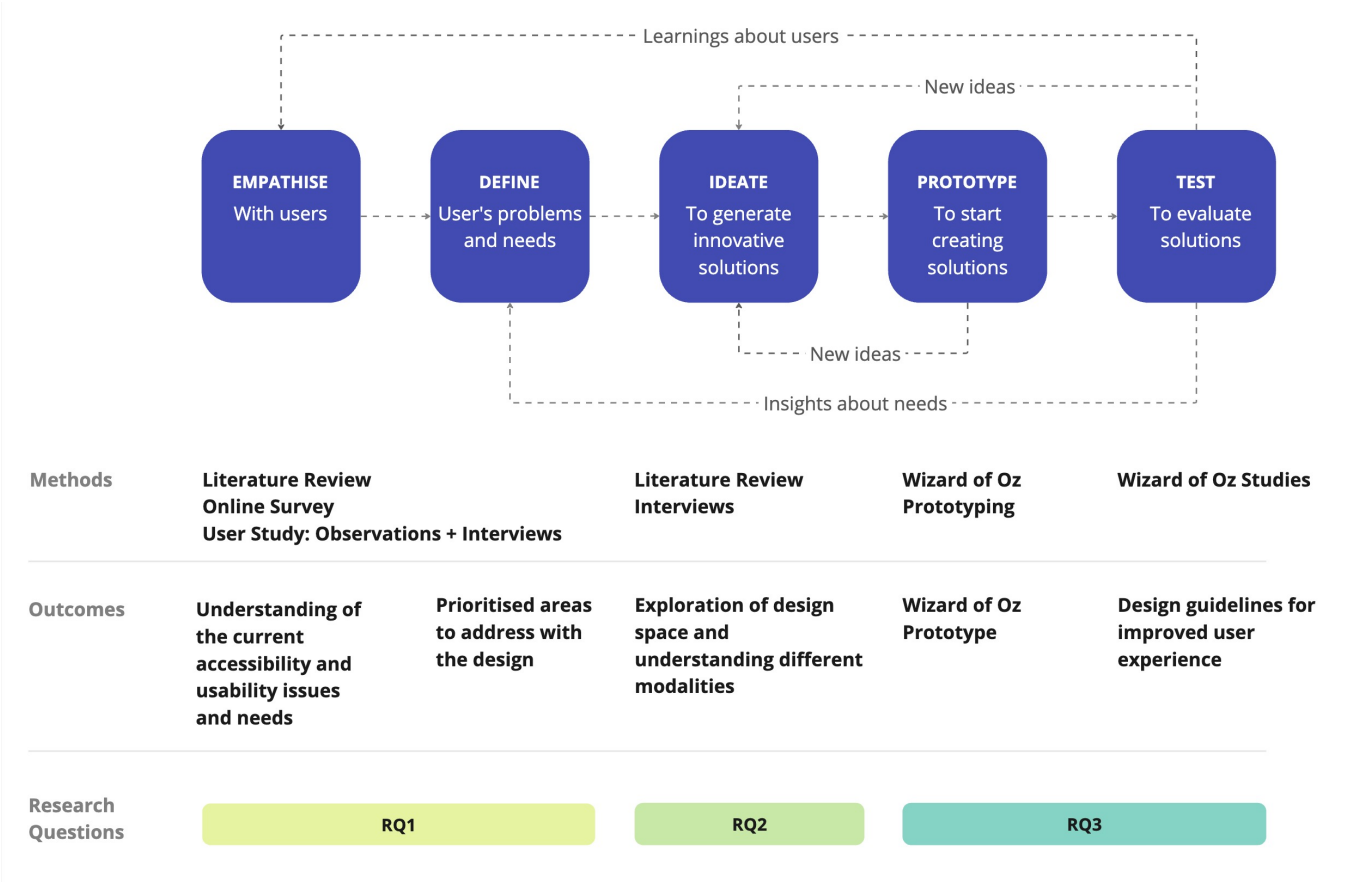


Figure 1: Overview of the methodological approach aligning to the research questions.

on users, their feelings when encountering these problems, and the strategies employed to overcome them. Lastly, we explored the potential use and role of AI technology in productivity applications. The survey was promoted online through mailing lists dedicated to individuals who are blind. Responses for open-ended questions were thematically analyzed using an inductive approach [12].

A total of 102 blind productivity application users responded to the survey. Our study reveals that accessibility challenges have a significant adverse impact on users’ productivity, independence, and emotional well-being. We found that presentation tools, word processors, and spreadsheets have more accessibility issues for blind users compared to other applications such as email, video conferencing platforms, cloud storage and calendars. Our findings also indicate a strong preference within the blind community for AI-based assistants in productivity applications.

5.2 Study Two: User study with blind screen reader users to understand their challenges in productivity applications

We constructed a user study around a real-world work scenario to gain an in-depth understanding of the challenges faced by blind

users when understanding and extracting information in productivity applications, as well as to understand their current workarounds and strategies. We chose spreadsheets as they consist of more advanced data analysis features. The study was conducted with 12 blind spreadsheet users who used screen readers to access spreadsheets. The participants were given multiple tasks to complete in spreadsheets, and we observed the steps and strategies they used to achieve the tasks.

The study was conducted remotely via Zoom, and participants used their personal computers and the screen reader of their choice. Participants were requested to share their screens and audio. The study consisted of four phases:

- Initial questionnaire: Participants were first asked questions about their demographics and use of spreadsheets.
- Exploration task: Participants were then presented with a real-world scenario that required the use of a spreadsheet in a workplace context. A spreadsheet was shared with participants, which contained multiple tables and data analysis features, including formatting and formulas. They were asked to provide an overview description of the spreadsheet, highlighting important information.

- Follow-up tasks: After exploring the spreadsheet the participants were asked specific questions about the data in the spreadsheet.
- Interviews: We then asked questions based on observations made during the exploration and follow-up tasks, primarily to have them share their experiences with navigation and use of features. We also asked them for suggestions to address the challenges experienced with spreadsheets.

The observations and interview responses were thematically analyzed using an inductive approach. We found that ensuring information equivalence and making features accessible does not necessarily guarantee usability for screen reader users. We identified five underlying challenges that had an impact on the experience of blind users, including cognitive overload, trade-offs between time and information, lack of awareness and expertise, insufficient system feedback, and absences in screen reader responses.

Two papers were submitted to CHI 2024 based on the findings of these studies.

6 NEXT STEPS

The first two studies emphasize the imperative for further research on the design of conversational AI-based assistants in productivity applications. While prior research [52] has explored document-centered assistance with conversational AI, it primarily focuses on sighted users. Building on our research progress, we plan to conduct interviews and Wizard of Oz studies with blind users to explore the design space of generative conversational AI in productivity applications.

6.1 Design Space including Interviews with Blind Users

During the Ideate phase (Figure 1), interviews with blind users will be conducted to explore the desired task assistance and interactions with AI assistants in productivity applications. We plan to build the conversations around the top three applications which were identified as having the most issues from the survey results. It is important to understand specific types of queries that blind users will formulate when seeking assistance as well as their expected responses from the assistant. As the applications are predominantly accessed using screen readers, the aim is to investigate the interactions between screen readers and conversational interfaces when employed within productivity applications. While previous research has explored voiced-based conversational interfaces and screen reader interactions [6, 42] in the web, there is a necessity of exploring this dynamic within the context of productivity applications.

The interviews are also expected to explore blind users' trust in AI assistants and preferred content verification mechanisms, users' level of task satisfaction and identify potential challenges that could arise with the use of conversational AI technologies in these applications.

6.2 Wizard of Oz Studies

Based on the findings from the interviews, Wizard of Oz prototypes will be designed utilizing publicly released generative conversational AI technologies such as ChatGPT or with recently introduced

AI assistants such as Microsoft Copilot [34]. These will be tested in the top three applications which had most accessibility issues around work related tasks. For example, creating a presentation on a topic, summarizing a table, describing a graphic. The Wizard of Oz prototype will be used to simulate system functionalities during the testing phase, allowing for the evaluation of user interactions. Wizard of Oz studies will be conducted through multiple iterations involving blind productivity application users. These studies will help gather feedback and insights on how users interact with the system. We will incorporate the insights gained from each iteration into the design and test them in subsequent iterations.

6.3 Focus for colloquium

The rapid growth of generative conversational AI potentially has broad applicability in the domain of accessibility for productivity applications. I hope to obtain valuable feedback from the CHI community, which will assist me in scoping functionalities or applications to address, establishing criteria for evaluation, and positioning my work in the field of accessibility.

7 CONTRIBUTIONS

When addressing the research questions the following contributions are anticipated:

- Better understanding of the accessibility and usability issues faced by blind users when working with productivity applications (RQ1)
- Exploration of design space on the level of assistance that blind users desire from AI assistants in effectively using productivity applications (RQ2)
- Better understanding of how AI assistants can be designed for more accessible and usable productivity applications for people who are blind (RQ3)

Higher productivity, resulting from the ability to work efficiently and independently with productivity tools without frustrations, can contribute to improved access to education and employability, ultimately enhancing the quality of life for people who are blind.

8 LONG TERM GOALS

My long-term goal is to contribute to accessibility research and work towards building a technology-based, inclusive work culture for people who are blind.

ACKNOWLEDGMENTS

I would like to thank my supervisors, Prof. Kim Marriott, Dr. Swamy Ananthanarayan, and Dr. Gagatay Goncu, for all the support and guidance. I would also like to express my appreciation to all the participants of the studies for their time and valuable feedback.

REFERENCES

- [1] Ali Abdolrahmani, Ravi Kuber, and Stacy M Branham. 2018. "Siri Talks at You" An Empirical Investigation of Voice-Activated Personal Assistant (VAPA) Usage by Individuals Who Are Blind. In *Proceedings of the 20th International ACM SIGACCESS Conference on Computers and Accessibility*. 249–258.
- [2] Ali Abdolrahmani, Kevin M Storer, Antony Rishin Mekkath Roy, Ravi Kuber, and Stacy M Branham. 2020. Blind leading the sighted: drawing design insights from blind users towards more productivity-oriented voice interfaces. *ACM Transactions on Accessible Computing (TACCESS)* 12, 4 (2020), 1–35.

- [3] Amazon. 2023. Alexa. Retrieved September 10, 2023 from <https://developer.amazon.com/alexa>
- [4] Apple. 2023. iWork. Retrieved September 10, 2023 from <https://www.apple.com/au/iwork/>
- [5] Apple. 2023. Siri. Retrieved September 10, 2023 from <https://www.apple.com/siri/>
- [6] Vikas Ashok, Yevgen Borodin, Yury Puzis, and IV Ramakrishnan. 2015. Captispeak: a speech-enabled web screen reader. In *Proceedings of the 12th International Web for All Conference*. 1–10.
- [7] Vikas Ganjigunte Ashok. 2018. *Non-Visual Web Browsing: From Accessibility with Screen Readers to Usability with Assistants*. Ph.D. Dissertation. State University of New York at Stony Brook.
- [8] Marcos Baez, Claudia Maria Cutrupi, Maristella Matera, Isabella Possaghi, Emanuele Pucci, Gianluca Spadone, Cinzia Cappiello, and Antonella Pasquale. 2022. Exploring challenges for conversational web browsing with blind and visually impaired users. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts*. 1–7.
- [9] Matthew J Baker, EM Nightingale, and Suzy Bills. 2021. An editing process for blind or visually impaired editors. *IEEE Transactions on Professional Communication* 64, 3 (2021), 275–287.
- [10] Stacy M Branham and Shaun K Kane. 2015. The invisible work of accessibility: how blind employees manage accessibility in mixed-ability workplaces. In *Proceedings of the 17th international acm sigaccess conference on computers & accessibility*. 163–171.
- [11] Stacy M Branham and Antony Rishin Mukkath Roy. 2019. Reading between the guidelines: How commercial voice assistant guidelines hinder accessibility for blind users. In *Proceedings of the 21st International ACM SIGACCESS Conference on Computers and Accessibility*. 446–458.
- [12] Virginia Braun and Victoria Clarke. 2012. *Thematic analysis*. American Psychological Association.
- [13] Maria Claudia Buzzi, Marina Buzzi, Barbara Leporini, Giulio Mori, and Victor MR Penichet. 2010. Accessing Google docs via screen reader. In *Computers Helping People with Special Needs: 12th International Conference, ICCHP 2010, Vienna, Austria, July 14–16, 2010. Proceedings 12*. Springer, 92–99.
- [14] Maitraye Das, Anne Marie Piper, and Darren Gergle. 2022. Design and evaluation of accessible collaborative writing techniques for people with vision impairments. *ACM Transactions on Computer-Human Interaction* 29, 2 (2022), 1–42.
- [15] Paul S Duckett and Rebekah Pratt. 2001. The researched opinions on research: Visually impaired people and visual impairment research. *Disability & Society* 16, 6 (2001), 815–835.
- [16] D Gareth Evans, Tim Diggle, Sri H Kurniawan, and Paul Blenkhorn. 2003. An investigation into formatting and layout errors produced by blind word-processor users and an evaluation of prototype error prevention and correction techniques. *IEEE Transactions on neural systems and rehabilitation engineering* 11, 3 (2003), 257–268.
- [17] Shubham Melvin Felix, Sumer Kumar, and A Veeramuthu. 2018. A smart personal AI assistant for visually impaired people. In *2018 2nd international conference on trends in electronics and informatics (ICOEI)*. IEEE, 1245–1250.
- [18] Shengjia Feng, Peter Buxmann, et al. 2020. My Virtual Colleague: A State-of-the-Art Analysis of Conversational Agents for the Workplace. In *HICSS*. 1–10.
- [19] Google. 2023. Google Assistant. Retrieved September 10, 2023 from <https://assistant.google.com/>
- [20] Google. 2023. Google Workspace. Retrieved September 10, 2023 from <https://workspace.google.com/>
- [21] Google. 2024. Bard. Retrieved January 19, 2024 from <https://bard.google.com/>
- [22] Tatsuya Ishihara, Hironobu Takagi, Takashi Itoh, and Chieko Asakawa. 2006. Analyzing visual layout for a non-visual presentation-document interface. In *Proceedings of the 8th international ACM SIGACCESS conference on Computers and accessibility*. 165–172.
- [23] Youssef Bou Issa, Mustapha Mohajid, Bernard Oriola, and Nadine Vigouroux. 2009. Accessibility for the blind: An automated audio/tactile description of pictures in digital documents. In *2009 International Conference on Advances in Computational Tools for Engineering Applications*. IEEE, 591–594.
- [24] Vinayak Iyer, Kshitij Shah, Sahil Sheth, and Kailas Devadkar. 2020. Virtual assistant for the visually impaired. In *2020 5th International Conference on Communication and Electronics Systems (ICCES)*. IEEE, 1057–1062.
- [25] Martin Jung and Erdmuthe Meyer zu Bexten. 2002. Visually impaired persons using Microsoft Powerpoint presentations. In *Computers Helping People with Special Needs: 8th International Conference, ICCHP 2002 Linz, Austria, July 15–20, 2002 Proceedings 8*. Springer, 463–468.
- [26] Everlyne Kimani, Kael Rowan, Daniel McDuff, Mary Czerwinski, and Gloria Mark. 2019. A conversational agent in support of productivity and wellbeing at work. In *2019 8th international conference on affective computing and intelligent interaction (ACII)*. IEEE, 1–7.
- [27] Cheuk Yin Phipson Lee, Zhuohao Zhang, Jaylin Herskovitz, JooYoung Seo, and Anhong Guo. 2022. Collaborally: Accessible collaboration awareness in document editing. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [28] Hae-Na Lee. 2023. *Enhancing the Usability of Computer Applications for People With Visual Impairments via UI Augmentation*. Ph.D. Dissertation. State University of New York at Stony Brook.
- [29] Hae-Na Lee, Vikas Ashok, and IV Ramakrishnan. 2021. Bringing things closer: Enhancing low-vision interaction experience with office productivity applications. *Proceedings of the ACM on Human-Computer Interaction* 5, EICS (2021), 1–18.
- [30] Irene Lopatovska, Katrina Rink, Ian Knight, Kieran Raines, Kevin Cosenza, Harriet Williams, Perachya Sorsche, David Hirsch, Qi Li, and Adrianna Martinez. 2019. Talk to me: Exploring user interactions with the Amazon Alexa. *Journal of Librarianship and Information Science* 51, 4 (2019), 984–997.
- [31] Ewa Luger and Abigail Sellen. 2016. "Like Having a Really Bad PA" The Gulf between User Expectation and Experience of Conversational Agents. In *Proceedings of the 2016 CHI conference on human factors in computing systems*. 5286–5297.
- [32] Akanksh A Manjunath, MC Sohan, MR Anala, and KN Subramanya. 2023. Design thinking approach to simplify monetary transactions for the people with visual impairment. *British Journal of Visual Impairment* 41, 2 (2023), 265–285.
- [33] Microsoft. 2023. Microsoft 365. Retrieved September 10, 2023 from <https://www.microsoft.com/en-au/microsoft-365>
- [34] Microsoft. 2023. Microsoft 365 Copilot. Retrieved November 9, 2023 from <https://www.microsoft.com/en-us/microsoft-365/enterprise/microsoft-365-copilot>
- [35] Lourdes Morales, Sonia M Arteaga, and Sri Kurniawan. 2013. Design guidelines of a tool to help blind authors independently format their word documents. In *CHI'13 Extended Abstracts on Human Factors in Computing Systems*. 31–36.
- [36] Giulio Mori, Maria Claudia Buzzi, Marina Buzzi, Barbara Leporini, and Victor MR Penichet. 2011. Making "Google Docs" user interface more accessible for blind people. In *Advances in New Technologies, Interactive Interfaces, and Communicability: First International Conference, ADNTIC 2010, Huerta Grande, Argentina, October 20–22, 2010, Revised Selected Papers 1*. Springer, 20–29.
- [37] Institute of Design at Stanford. 2023. An Introduction to Design Thinking PRO-CESS GUIDE. Retrieved November 9, 2023 from <https://web.stanford.edu/~mshanks/MichaelShanks/files/509554.pdf>
- [38] Liliana E Olguin-Gil, Francisco Vázquez-Guzmán, Eduardo Vázquez-Zayas, Jezreel Mejía, and Itzel Blanco-Cruz. 2022. Virtual Assistant as Support for People Visually Impaired. In *New Perspectives in Software Engineering: Proceedings of the 10th International Conference on Software Process Improvement (CIMPS 2021)*. Springer, 174–188.
- [39] Juliana Damasio Oliveira and Rafael H Bordini. 2019. A survey on the needs of visually impaired users and requirements for a virtual assistant in ambient assisted living. In *16th International Conference on Information Technology-New Generations (ITNG 2019)*. Springer, 449–453.
- [40] OpenAI. 2024. ChatGPT. Retrieved January 19, 2024 from <https://chat.openai.com/>
- [41] Sassy Outwater-Wright. 2020. Putting an end to the idea of working twice as hard: How Leadership Can Affect Change in the Workplace for People Who Are Blind. *Journal of Visual Impairment & Blindness* 114, 1 (2020), 70–72.
- [42] Mahika Phutane, Crescentia Jung, Niu Chen, and Shiri Azenkot. 2023. Speaking with My Screen Reader: Using Audio Fictions to Explore Conversational Access to Interfaces. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–18.
- [43] Alisha Pradhan, Kanika Mehta, and Leah Findlater. 2018. "Accessibility Came by Accident" Use of Voice-Controlled Intelligent Personal Assistants by People with Disabilities. In *Proceedings of the 2018 CHI Conference on human factors in computing systems*. 1–13.
- [44] Emanuele Pucci, Isabella Possaghi, Claudia Maria Cutrupi, Marcos Baez, Cinzia Cappiello, and Maristella Matera. 2023. Defining Patterns for a Conversational Web. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [45] Elizabeth B-N Sanders and Pieter Jan Stappers. 2008. Co-creation and the new landscapes of design. *Co-design* 4, 1 (2008), 5–18.
- [46] Sergio Sayago and Mireia Ribera. 2020. Apple Siri (input)+ Voice Over (output)= a de facto marriage: An exploratory case study with blind people. In *Proceedings of the 9th International Conference on Software Development and Technologies for Enhancing Accessibility and Fighting Info-exclusion*. 6–10.
- [47] Anastasia Schaadhardt, Alexis Hiniker, and Jacob O Wobbrock. 2021. Understanding blind screen-reader users' experiences of digital artboards. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems*. 1–19.
- [48] John Gerard Schoeberlein and Yuanqiong Wang. 2014. Usability Evaluation of an Accessible Collaborative Writing Prototype for Blind Users. *Journal of Usability Studies* 10, 1 (2014).
- [49] AM Silverman, LP Rosenblum, EC Bolander, CR Rhoads, and K Bleach. 2022. Technology and accommodations: Employment experiences of US adults who are blind, have low vision, or are deafblind. American Foundation for the Blind.
- [50] Tony Stockman. 2004. The design and evaluation of auditory access to spreadsheets. In *ICAD 04-Tenth Meeting of the International Conference on Auditory Display*.
- [51] Tony Stockman, Christopher Frauenberger, and Greg Hind. 2005. Interactive sonification of spreadsheets. In *ICAD 05-Eleventh Meeting of the International Conference on Auditory Display*.

- [52] Maartje ter Hoeve, Robert Sim, Elnaz Nouri, Adam Fourney, Maarten de Rijke, and Ryen W White. 2020. Conversations with documents: An exploration of document-centered assistance. In *Proceedings of the 2020 Conference on Human Information Interaction and Retrieval*. 43–52.
- [53] Zhuohao Zhang and Jacob O Wobbrock. 2023. A11yBoard: Making Digital Artboards Accessible to Blind and Low-Vision Users. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [54] Zhuohao Jerry Zhang, Gene SH Kim, and Jacob O Wobbrock. 2023. Developing and Deploying a Real-World Solution for Accessible Slide Reading and Authoring for Blind Users. In *Proceedings of the 25th International ACM SIGACCESS Conference on Computers and Accessibility*. 1–15.